

Time Series — Solutions

Practice Exam — Week 1: Stationarity, Autocovariance & Ergodicity
EPFL, MATH-342 · Prof. S. C. Olhede

Problem 1 (5 points)

(a) $\{X_t\}$ is *weakly (second-order) stationary* if all first- and second-order moments are finite and shift-invariant: $\mathbb{E}[X_t] = \mu$ (constant), $\text{Var}(X_t) = \sigma^2 < \infty$ (constant), and $\text{Cov}(X_t, X_{t+\tau})$ depends on the lag τ only (not on t). It is *strictly stationary* if for every n and all $t_1, \dots, t_n$, the joint law of $(X_{t_1}, \dots, X_{t_n})$ equals that of $(X_{t_1+\tau}, \dots, X_{t_n+\tau})$ for all τ .

(b)(i) $\mathbb{E}[X_t] = a$. By independence of the Y_t ,

$$\gamma_\tau = \begin{cases} (b^2 + c^2)\sigma_Y^2, & \tau = 0, \\ bc\sigma_Y^2, & \tau = \pm 1, \\ 0, & |\tau| \geq 2. \end{cases}$$

Depends on τ only and is finite $\Rightarrow$ weakly stationary; a linear combination of jointly Gaussian variables is Gaussian $\Rightarrow$ also strictly stationary.

(b)(ii) $\mathbb{E}[X_t] = 0$ but $\text{Cov}(X_t, X_{t+\tau}) = \sigma_Y^2 \cos(ct) \cos(c(t+\tau))$ depends on $t \Rightarrow$ not stationary (neither weakly nor strictly).

(c) An i.i.d. sequence of Cauchy (Student- t_1) variables: i.i.d. hence strictly stationary, but it has no finite mean/variance, so the finite-variance requirement of weak stationarity fails. Thus strict $\not\Rightarrow$ weak.

Problem 2 (5 points)

(a) Independence kills all cross-covariances $\text{Cov}(X_s, \varepsilon_u) = 0$, and the white noise contributes variance only at lag 0:

$$\gamma_\tau^Z = \gamma_\tau + \sigma^2 \mathbf{1}_{\{\tau=0\}}.$$

(b) By bilinearity and $\text{Cov}(\varepsilon_s, \varepsilon_u) = \sigma^2 \mathbf{1}_{\{s=u\}}$ (only equal noise indices survive),

$$\gamma_\tau^W = \begin{cases} (\theta_1^2 + \theta_2^2)\sigma^2, & \tau = 0, \\ \theta_1\theta_2\sigma^2, & \tau = \pm 1, \\ 0, & |\tau| \geq 2, \end{cases} \quad \rho_1 = \frac{\theta_1\theta_2}{\theta_1^2 + \theta_2^2}.$$

By AM–GM, $|\theta_1\theta_2| \leq \frac{1}{2}(\theta_1^2 + \theta_2^2)$, hence $|\rho_1| \leq \frac{1}{2}$, with equality iff $|\theta_1| = |\theta_2|$. (No process of this MA(1)-type can exceed $|\rho_1| = \frac{1}{2}$.)

Problem 3 (5 points)

(a) Let $W = \sum_{j=1}^n a_j X_j$. A variance is non-negative, so

$$0 \leq \text{Var}(W) = \sum_{j=1}^n \sum_{k=1}^n a_j a_k \text{Cov}(X_j, X_k) = \sum_{j=1}^n \sum_{k=1}^n a_j a_k \gamma_{j-k}.$$

(b) Symmetry: $\gamma_{-\tau} = \text{Cov}(X_t, X_{t-\tau}) = \text{Cov}(X_{t-\tau}, X_t) = \gamma_\tau$ (real case). Bound: by Cauchy–Schwarz $|\gamma_\tau| = |\text{Cov}(X_t, X_{t+\tau})| \leq \sqrt{\text{Var}(X_t) \text{Var}(X_{t+\tau})} = \gamma_0$.

(c) **No.** A valid ACVS must be positive semi-definite, equivalently have a non-negative spectral density. Here

$$\mathcal{S}(f) = \sum_{\tau} \gamma_{\tau} e^{-2\pi i f \tau} = 1 + 1.6 \cos(2\pi f),$$

which equals $1 - 1.6 = -0.6 < 0$ at $f = \frac{1}{2}$. The spectral density goes negative $\Rightarrow$ the sequence is not p.s.d. $\Rightarrow$ not a valid ACVS. (Already $|\rho_1| = 0.8 > \frac{1}{2}$ is impossible for an MA(1).)

Problem 4 (5 points)

(a) $\tilde{\gamma}_{\tau}$ is *unbiased* (known mean) but not always positive semi-definite; $\hat{\gamma}_{\tau}$ is *biased* ($\mathbb{E}[\hat{\gamma}_{\tau}] = \frac{n-|\tau|}{n} \gamma_{\tau}$) but positive semi-definite — which is why it is the default.

(b) For white noise both are unbiased, so $\text{MSE} = \text{Var}$. Now

$$\text{Var}(\hat{\gamma}_{\tau}) = \frac{n-|\tau|}{n^2} \sigma^4 = \left(\frac{n-|\tau|}{n} \right)^2 \frac{\sigma^4}{n-|\tau|} = \left(\frac{n-|\tau|}{n} \right)^2 \text{Var}(\tilde{\gamma}_{\tau}) \leq \text{Var}(\tilde{\gamma}_{\tau}).$$

So the *biased* estimator $\hat{\gamma}_{\tau}$ has the smaller MSE. (Lesson: the goal is low MSE and positive semi-definiteness, not unbiasedness.)

(c) $\mathbb{E}[\bar{X}] = \mu$, and grouping by lag,

$$\text{Var}(\bar{X}) = \frac{1}{n^2} \sum_{i,j=1}^n \gamma_{j-i} = \frac{1}{n} \sum_{\tau=-(n-1)}^{n-1} \left(1 - \frac{|\tau|}{n} \right) \gamma_{\tau}.$$

By Cesàro summability, $\sum_{\tau} \left(1 - \frac{|\tau|}{n} \right) \gamma_{\tau} \rightarrow \sum_{\tau=-\infty}^{\infty} \gamma_{\tau} =: C(\gamma)$, finite since $\sum_{\tau} |\gamma_{\tau}| < \infty$. Hence $n \text{Var}(\bar{X}) \rightarrow C(\gamma)$, i.e. $\text{Var}(\bar{X}) = O(1/n) \rightarrow 0$, so $\bar{X} \rightarrow \mu$ in mean square, therefore in probability: $\bar{X}$ is consistent (the process is mean-ergodic).